



10160 - The Ancient Star Formation History of a Rare Isolated Ultra-faint Dwarf Galaxy

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Sculptor A				
	1	SculptorA	NIRCam Imaging	(1) SCULPTOR-A

ABSTRACT

Ultra-faint dwarf galaxies are the faintest galaxies in the universe and represent an extreme of galaxy properties. Based on their star formation histories (SFHs), these systems are thought to be 'fossil' galaxies that began forming stars at early epochs only to be quenched within a few Gyr. Their arrested growth is typically attributed to the combined effects of stellar feedback and reionization.

The recently discovered ultra-faint dwarf Sculptor A challenges this prevailing framework. Sculptor A is an isolated ultra-faint dwarf, at the edge of the Local Group (1.1 Mpc) and $>3 R_{\text{vir}}$ from a massive galaxy. With $M_{\text{star}} = 6.3 \times 10^4 M_{\text{sun}}$, Sculptor A is below the mass threshold where reionization is expected to quench galaxies. However, shallow HST imaging suggests that Sculptor A has an extended SFH. If confirmed, this would suggest that (i) the characteristic mass scale for reionization quenching is lower than current models predict; (ii) environment may have played a role in quenching ultra-faint dwarfs of a similar mass that are satellites (i.e., within the virial radius of a massive host). We propose deep NIRCам imaging of Sculptor A to derive the ancient SFH of the galaxy. The results will confirm whether Sculptor A has an extended SFH (in which case it will constrain the timescale for slower environmental quenching) or if it experienced rapid quenching by reionization. This will provide a new empirical benchmark for simulations on the galaxy mass scale susceptible to quenching via reionization. Sculptor A's isolation and proximity provide a rare opportunity to disentangle the effects of reionization and environment on small-scale structure with minimal JWST time.

OBSERVING DESCRIPTION

We will image the galaxy with NIRCам with the short-wavelength filters F090W and F150W, with simultaneous imaging in the long-wavelength filters F480W and F360M. Our science goals require that the SW imaging reach below the old main sequence turn-off with a $\text{SNR} \sim 10$ for a metal-poor, 13 Gyr old star. We have used a MEDIUMDEEP8 readout and a subpixel dither pattern to mitigate cosmic rays and bad pixels, improve PSF sampling, and reduce flat-field errors thereby maximizing the SNR. We do not intend to cover the chip gaps as uniform depth is preferred for the imaged stellar fields. We have selected 6-7 groups per integration. Based on the JWST ETC v5 and results from the ERS program on resolved stellar populations, a total science time of 7.7 hrs is needed to achieve our photometric depth requirement. Total requested time with overhead from the APT is 10.1 hours. We impose a background limited constraint of 10% for the galaxy as the background is highly variable in this region of the sky. We selected additional orient constraints to ensure the NIRCам field of view optimally covers the stellar disk of the galaxy while avoiding any bright nearby stars and the meteoroid zone.

Proposal 10160 - Targets - The Ancient Star Formation History of a Rare Isolated Ultra-faint Dwarf Galaxy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SCULPTOR-A	RA: 01 08 30.8200 (17.1284167d) Dec: -36 03 56.30 (-36.06564d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Dwarf galaxies]				

Proposal 10160 - Observation 1 - The Ancient Star Formation History of a Rare Isolated Ultra-faint Dwarf Galaxy

Observation	Proposal 10160, Observation 1: SculptorA									Thu Jun 11 18:56:52 GMT 2026		
	Diagnostic Status: Warning											
	Observing Template: NIRCcam Imaging											
	Comments: We request that these observations not be taken directly following observations of a very bright target. From the ERS program, persistance can impact the precise photometric recovery of faint stars, which is the primary goal of this observing program (Weisz et al. 2023).											
Diagnostics	(SculptorA (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous				
	(1)	SCULPTOR-A	RA: 01 08 30.8200 (17.1284167d) Dec: -36 03 56.30 (-36.06564d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Dwarf galaxies]											
Template	Module		Subarray				Target Placement					
	ALL		FULL				Module gap (large extended source)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions		
	1	NONE				STANDARD				4		
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F090W	F360M	MEDIUMDEEP8	7	4	16	4	16964.097			
	2	F150W	F480M	MEDIUMDEEP8	6	3	12	4	10779.717			
Special Requirements	Aperture PA Range 250 to 35 Degrees (V3 250.07457694 to 35.07457694) Offset -119.0 arcsec, -34.0 arcsec Background Limited. Background no more than 10th percentile above minimum											